An Epilepsy Prediction Approach Based on Semi-supervised Deep Convolutional Generative Adversarial Nets

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Abstract: Epilepsy is a chronic disease caused by abnormal brain discharge leading to transient brain dysfunction. In recent years, studies based on EEG to achieve epilepsy prediction have attracted more and more attention. This paper proposes an epilepsy prediction framework based on a semi-supervised deep convolutional generated adversarial network (SS-DCGAN). In this framework, we choose partial samples to train model and makes the model more stable by adding constraints to the loss function. The model adjusts the loss function of the discriminator and uses the sample tag for the construction of supervised losses, to achieve the purpose of semi-supervised training. The approach is validated on the kaggle dataset and the chbmit dataset. Experiments show that the model achieves better results on the kaggle dataset, with an average classification accuracy of 95.54% and an average accuracy of 93.67% on the chbmit dataset.

1. Introduction

Epilepsy is a disease caused by abnormal neuronal activity, with approximately 50 million epilepsy patients worldwide [1] [2]. Among them, only two-thirds of patients can receive effective treatment, and only 7-8% of those can be cured by surgery [3]. Loss of consciousness, body stiffness or convulsions are the main symptoms of seizures [4], which may make the patient or the public in danger. Although a variety of anti-epileptic drugs have been available in the past 20 years, the proportion of patients with refractory epilepsy has not decreased.

EEG signals are often used to determine the onset and type of epilepsy [5] [6]. There are two types EEG signals, including scalp EEG signals and intracranial EEG signals, clinically. The scalp EEG signal is captured by a wearable device such as an electrode cap, and the damage to the patient is smaller than that of the intracranial EEG signa. In recent years, researches on epilepsy prediction have attracted more and more attention. Most of researches based on EEG signal focus on machine learning methods.

At present, machine learning research methods mainly include: supervised learning, unsupervised learning, semi-supervised learning, and reinforce learning [7]. Many methods are based on supervised training, such as support vector machine (SVM), decision tree (DT), convolutional neural network (CNN), and so on. R. Shantha Selvakumari et al. combined the principal component analysis (PCA) and Poincaré section to abstract feature, and used a classifier consisting of SVM and naive Bayes (NB) to determine seizures or not[8]. In order to predict epilepsy more accurately, Samed Jukic et al. remove signal noise by multi-scale principal component analysis (MSPCA), and classify the features obtained by wavelet packet decomposition using rotating forest (RF) [9]. Ye Yuan et al. introduced channel-aware techniques based on attention-based decision-making methods into the detection of epilepsy, using a multi-view deep learning framework to achieve dynamic determination of critical electrodes and seizure detection [10]. Nhan Duy Truong et al. used the short-time Fourier transform (STFT) and CNN to achieve the goal of epilepsy prediction [12]. Tsiouris et al. extracted cross-correlation

features, time-frequency features, and used graph theory knowledge and long-short-term memory network (LSTM) to perform epilepsy prediction tasks for 5s EEG signals [13].
In the field of epilepsy research, most EEG data are unlabeled, and labeling EEG is very costly, so unsupervised learning and semi-supervised learning methods have also received extensive attention. Orhan U et al. use the KMeans algorithm to cluster the discrete wavelet transform results of each sub-band, and then obtain the probability distribution according to the clustering result of the wavelet coefficient cluster, which as the input of the multi-layer perceptron to distinguish normal human and patients with epilepsy [14]. In 2017, Li Dongmei et al proposed an multi-lead EEG data classification and epileptogenic localization approach based on PCA. Combined with PCA and nonlinear dynamics, the approach used random forest algorithm toachieve accurate classification of EEG signals and further pathogen localization [15]. As an unsupervised learning method, the self-encoder is often used to perform feature extraction of EEG signals, recently. In order to reduce the influence of myoelectric artifacts on epilepsy detection, Qi Y et al. proposed a robust stacked autoencoder (R-SAE) to extract effective features [16]. Barata et al. also applied SAE to the feature extraction of epilepsy prediction, and the proposed prediction method performed well on long-term EEG signals collected by 103 patients [17]. Belhadj et al. extracted features of EEG signal using burg method, and used them as inputs to a sparse self-auto-encoder (SAE), and finally used SVM to perform epilepsy detection [18]. As a unsupervised learning method, Generative Adversarial Nets(GAN) has evolved into a variety of different models [19].

Semi-supervised learning is a method between supervised learning and unsupervised learning. It explores regular pattern based on a small number of labeled samples and a large number of unlabeled samples. In addition to the semi-supervised model derived from the generative model such as GAN, there are quite a few semi-supervised models based on methods such as deep belief network (DBN) and SAE. Wulsin D F et al. proposed a semi-supervised DBN model and applied the model to achieve abnormal detection of epileptic signals [22]. Gogna A et al. modified the structure of the stacked self-encoder to meet the requirements of signal compression and classification [23]. Jiang Y et al. integrated migration learning, semi-supervised learning, and TSK fuzzy systems to accomplish the task of identifying seizures [24]. Singhal V et al. proposed a new semi-supervised deep CS technique that combines deep learning analysis capabilities with compressed sensing (CS) reconstruction capabilities, which yields better classification results on the Bonn dataset [25].

As an unsupervised method, DCGAN method has certain advantages in feature learning, but there are still some problems in the optimization direction constraint, which may result in the model may not meet the requirements. To this end, this paper proposes a semi-supervised deep convolutional generative adversarial nets (SS-DCGAN) to accomplish epilepsy prediction. Based on DCGAN, this approach obtains more stable model through adding labeled data and constraints .

2. Related Approach

2.1 GAN

GAN is originated from game theory. It consists of two parts, including Discriminative Model (D)

and Generative Model (G). The optimization of the two is alternated. The network model is trained by means of adversarial learning. To achieve the task of learning features from the original sample and generating new samples [26].

The processing flow of GAN is depicted as follows. Assuming that the real data X conforms to the distribution $P_{data}(x)$, the noise data z conforms to the distribution $P_{\text{noise}}(z)$. The noise data is used as an input to generation model G, and the generated data G(Z) is obtained, which is subject to distribution $P_g(G(\text{z}))$. Then, the generated data and real data are input to the discriminant model D. In the early stage of training, the discriminator can easily determine the generated data and the real data, due to lack of prior knowledge and feedback information. After several rounds of alternating training, the generator produces data that is more difficult to distinguish between true and false, and the discriminator gains stronger discriminating power. The optimization objective function of GAN is:

$$L = \arg\min_{G}\max_{D} V(G, D)$$

$$V(G, D) = \mathbb{E}_{x\sim P_{data}(\text{x})}[\log D(x)] + \mathbb{E}_{z\sim P_{\text{noise}}(\text{z})}[\log(1 - D(G(z)))]$$

GAN has the advantages of strong versatility of training methods and complex calculations to avoid approximate probability. However, the model also has some defects. For example, generator's crashes will make the training process stop, and resulting model may not meet the requirement due to lack of constraints.

2.2 DCGAN

With the help of CNN's advantages in image feature extraction, DCGAN enhances learning effect through optimizing GAN training and adding constraints. The basic component discriminator and generator have been modified on the basis of GAN, including:

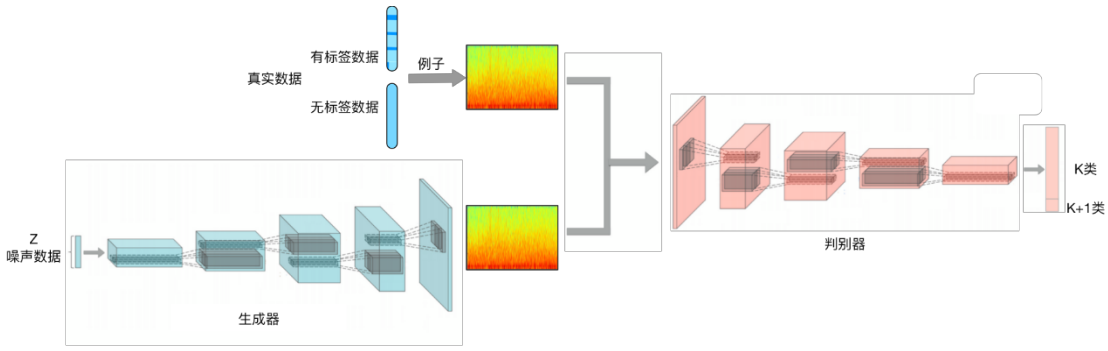
- Combining CNN and GAN.
- Making the training process more stable by setting a series of constraints on network's configuration.
- Demonstrating the vector calculation characteristics of the generated feature representation by qualitatively analysis of the filter learned from the GAN network.
- Replacing the pooling layer and the convolution layer function with the fragmented convolutions and fractional-strided convolutions in the discriminator and generator sections, respectively.

In order to solve the problems of gradient explosion, disappearance, slow convergence, etc. caused by improper initialization, and make the initialization effect more stable. The approach applies Batch Normalization in each layer, except input layer, in generative model and discriminator model. This strategy helps propagate gradient to each layer, and prevents the generator from converge to the same point due to improper initialization [27]. In addition, the model adds a global pooling operation based on the removal of the fully connected layer, which reduces the convergence speed of the model and increases the stability of the model. Since the use of bounded functions is helpful for the model to quickly cover the color space in the training distribution, the model also changes the use of the activation function, including:

- Applying Relu as the activation function for each layer in generator, except the output layer.
- Applying leaking Relu as the activation function for each layer in discriminator.

3. Our Approach

This paper proposes a framework for epilepsy prediction based on SS-DCGAN. As shown in Fig. 1, the framework adds the training process and constraint direction of model training by samples' label, while preserving DCGAN's advantage in image feature learning. By this method, a model that performs well in both feature learning and classification is obtained, and the model distinct between the pre- and post-stage of epileptic EEG signals.



SS-DCGAN modifies its discriminant model based on the original DCGAN. The function of the original DCGAN generation model G remains unchanged, and the function of the discriminator D is extended by adding function of judging the real sample label to the classifier. The discriminator not only distinguish the real image from the generated images, also subdivides the category into K+1 class, which includes K category of the real image and the unique label of the generated image. The approach also modified the loss function of DCGAN to achieve the purpose of model training, through the joint constraint of supervised loss and unsupervised loss.

The input of the discriminator consists of two parts: labeled data x_l and unlabeled data x_u, wherein the unlabeled data is derived from the generator sampling from the random noise z, generated pseudo samples x_g and partially unlabeled real data. The classifier simultaneously receives the original training data and the dummy data generated by the generator. The output includes the discrimination of true or false of data and the determinant of the data category. Finally, the corresponding probability value p is obtained by softmax, and the formula is as follows:

$$softmax(h_i) = \left(\frac{exp\ (h_i)}{\sum_j exp\ (h_j)}\right) \tag{3-1}$$

h_i represents the i-th output of forward layer of softmax. Through normalization operation on softmax, we can obtain a probability value p rang over [0, 1], and the maximum of p indicates corresponding label y. Model optimization can be achieved by maximizing $\log_{p_{model}}(y = K + 1|x)$ for unlabeled data when designing loss functions.

Discriminator D's training data includes real data and generated data, and its loss function consist of two parts: loss of supervised learning and loss of semi-supervised learning, and the formula is as the following:

$$Loss = -\mathbb{E}_{x,y\sim p_{data}(x,y)}\left[log_{p_{model}}(y|x)\right] - \mathbb{E}_{x\sim G}\left[log_{p_{model}}(y = K+1|x)\right]$$

$$= Loss_{\text{supervised}} + Loss_{\text{unsupervised}} \quad (3\text{-}2)$$

Loss applies cross entropy loss function. In the formula, $Loss_{supervised}$, $Loss_{unsupervised}$ represents the loss of supervised learning and the loss of unsupervised learning, respectively, x, y~$P_{(x,y)}$ and x~G represents the distribution of real data and generated data, respectively, in which, x is input sample, y represents related sample's tag, K represents the truly category of sample, and P_{model} represents model's distribution.

$$Loss_{supervised} = -\mathbb{E}_{x,y\sim p_{data}(x,y)}\left[log_{p_{model}}(y|x, y < K+1)\right] \quad (3\text{-}3)$$

$$Loss_{unsupervised} = -\left\{\begin{matrix}\mathbb{E}_{x,y\sim p_{data}(x)} log[1 - p_{model}(y = K+1|x)] \\ +\mathbb{E}_{x\sim G} log[p_{model}(y = K+1|x)]\end{matrix}\right\} \quad (3\text{-}4)$$

The task of unsupervised learning is determining whether the input data is real data or generated data. Let the probability of outputting a real data be D(x), which has the following formula:

$$D(x) = 1 - p_{model}(y = K+1|x) \quad (3\text{-}5)$$

Generated data is generated according to the noise z, which is subject to the distribution $P_{\text{noise}}(z)$, so the loss of unsupervised learning can be depicted as the following formula:

$$Loss_{unsupervised} = -\left\{\mathbb{E}_{x\sim p_{data}(x)} log[D(x)] + \mathbb{E}_{z\sim p_{noise}(z)} log\left[1 - D\left(G(z)\right)\right]\right\} \quad (3\text{-}6)$$

In formula 3-6, G(z) is generated data, which represents output result of generator G when the input is z. $Loss_{unsupervised}$ and the loss of GAN's discriminator have the same optimization target, which is finding the difference between real data and generated data, and promoting the distinguishing ability of GAN's discriminator. In the semi-supervised optimization process , the classifier of the discriminator is divided into one more class, which represents false data generated by generator, than the original class. In addition, the loss function of the discriminator includes supervised losses and unsupervised loss function that must be minimized during the training process.

Applying semi-supervised learning, discriminator obtain prior knowledge according to determination of false data and real data, and abstract some initial features, and continue to abstract feature and training model iteratively.

4. Experiments and Analysis

4.1 Experimental details

This paper uses the kaggle dataset and the epilepsy dataset provided by Boston Children's Hospital to accomplish the experiment. The partial sample information of the kaggle dataset is shown in

Table 1 epilepsy data of five groups of dogs

Detail info. / object	inter-	pre-	Data channels	frequency
Dog_1	480	240	16	400
Dog_2	500	420	16	400
Dog_3	1440	720	16	400

Dog_4	804	970	16	400
Dog_5	450	300	15	400

Table 1. The dataset is collected using a sampling frequency of 400 Hz and the reference electrode is an extracranial electrode. The data was divided into multiple 10-minute segments and labeled as pre-seizure or interictal. The data segments are numbered in chronological order and the test data is randomly arranged. In order to avoid data interference in the pre-, inter-, post-, and post-epileptic stages of epilepsy, no other seizures may be required within one week before and after data collection. The entire monitoring process is less than one week, and the inter-phase data collection time is strictly controlled at least four hours before or after the seizure, and is randomly selected from the subjects if the constraints are met.

The CHBMIT epilepsy dataset included multi-electrode scalp EEG signals in 18 women and 5 male epilepsy patients between the ages of 2 and 22, and the signals were collected on a device with a sampling frequency of 256 Hz. Most of the data are 23-channel EEG signals, and there are related annotations of time information such as seizures. The dataset contains more than 900 hours of EEG data, with a total of 198 seizures, with a total onset duration of approximately 190 minutes.

The data dividing method of reference [28] cuts multi-channel data under a consistent standard. The EEG data within 1 hour before the seizure was used as the pre-data, and the interictal data was separated from the two episodes by at least 4 hours, and the collected pre- and inter-phase data were cut into 10-minute data segments.

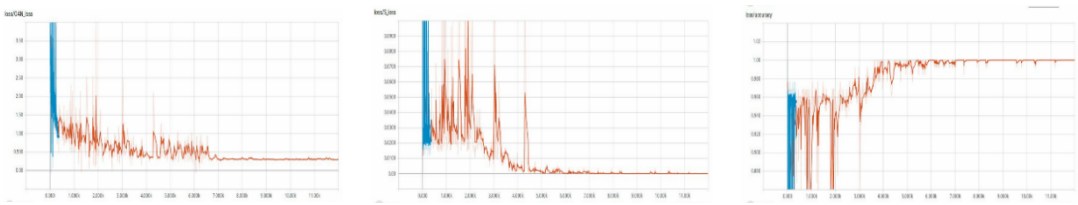
A short-time Fourier transform was performed on each channel of the multi-channel epileptic EEG data using a sliding window of 4 s to obtain a spectrogram with an overlap size of 2 s. The spectrum map and the corresponding label value are used as input of SS-DCGAN. The usage parameters of the network model are as shown in Table 2. In the training, a strategy such as dropout is used to prevent over-fitting.

Table 2 parameter setting

Parameter name	value	Parameter name	value
Batch size	128	Decay steps	10000
Learning rate	1e-4	Decay rate	0.5
Update rate	5	Dropout	0.5

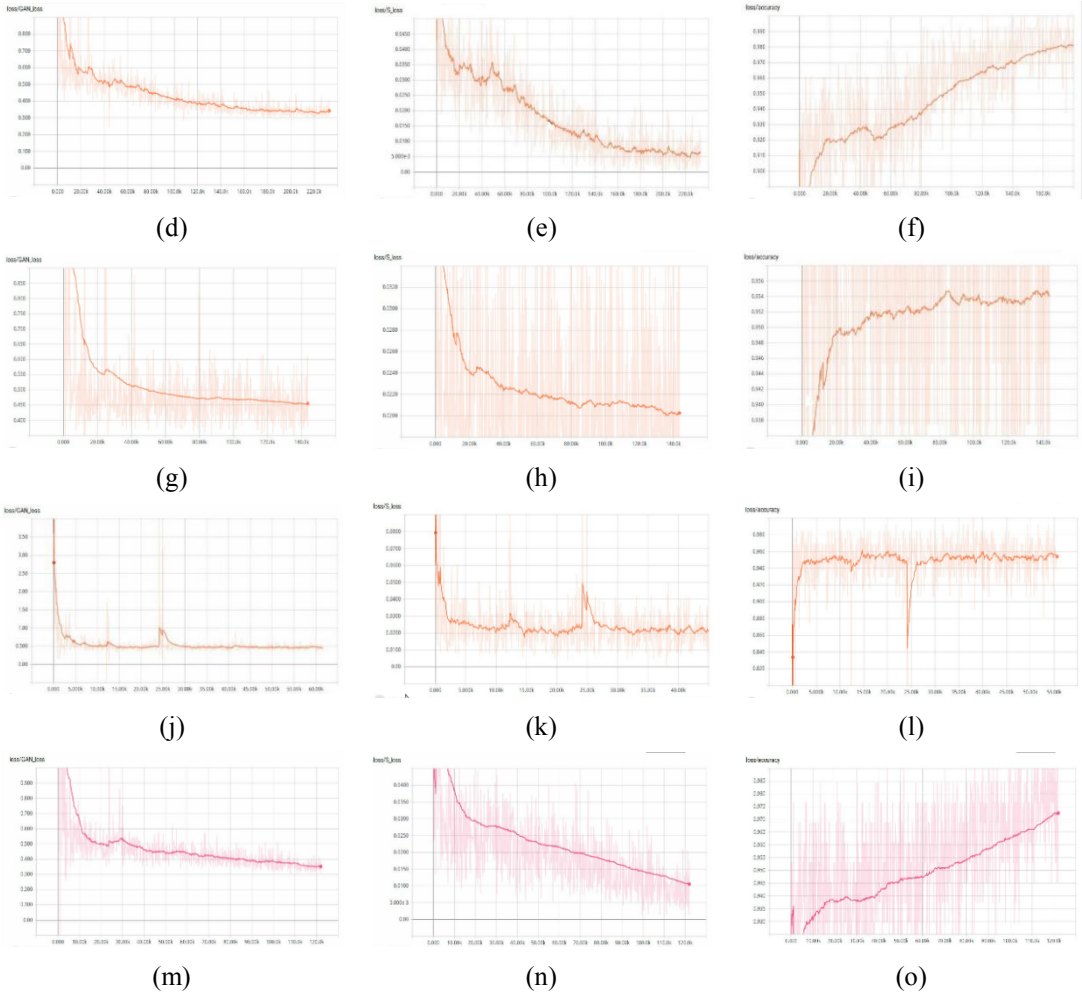
4.2 Experimental analysis

We carried out experiments on the CHBMIT dataset. The data was obtained from the EEG data of 11 patients. The signals collected by these patients contained a large amount of 23 channels of data, and the spectra were obtained according to the 10-minute data segment of each channel. These spectra are divided into training and test sets for training and testing of semi-supervised models. Since the amount of inter-stage data is much larger than the pre-stage data, the inter-phase data is randomly grouped, and the number of samples in each group is equal to the amount of pre-data. Before each training and test, a group is selected from the interval and all the



(a) (b) (c)

pre-samples are combined into a sample set, and then the training set and the test set are divided into the sample set to meet the training and evaluation requirements of the model. The process is repeated multiple times. Each operation selects a different subset of interval data, and the average accuracy obtained is 93.668%.



The specific loss change and accuracy curve of the model in the experimental process of kaggle epilepsy EEG data are shown in Figure 2. From top to bottom, the intermediate results are from Dog1 to Dog5, the loss of SS-DCGAN, GAN_loss and the supervised loss S_loss are presents from left to right, respectively. The display for each group of participants only contained supervised learning losses, unsupervised learning loss and accuracy. In the training process, the statistics are: the generator loss, the supervised loss of the discriminator for the tagged data, and the unsupervised loss of the discriminator to the generated data.

As shown in Fig. 3(a)-3(c), the training data of Dog1 is divided into two parts, the blue and orange parts are shown on the figure, and the blue part shows the process of the first 500 steps. It can be seen that the previous model was severely oscillated. In the later period, there is a gradual upward trend and tends to be stable, and basically reaches a steady state by about 7000 steps.

Dog2's classification accuracy rate curve trains to about 180,000 and basically achieves a stable maximum accuracy of 98%. The training process of Dog2 is relatively smooth, but because of the large number of Dog2 data and the slow optimization speed, the training process is longer and the number of iteration steps is more. A total of 487 steps were performed during the Dog2 test, and the average training accuracy was 98.1525%. During the training process, Dog3 and Dog5 have a

continuous decline in supervised loss, and the shock is more severe. The test results have higher classification accuracy than other methods, and the indicators on Dog5 are increased by more than 5%.

In order to verify the effectiveness of the our proposed method, this paper compares with some classical feature extraction methods. The results are shown in Table 3. The table contains Coherence mean (CM), Spectral entropy (SE), and Spectral relation power (SRP). As well as the classification results of the three combined features (splice), the paper also categorizes the results of directly classifying the spectral features, including CNN and ELM. In addition, in order to verify the effectiveness of the semi-supervised model, we directly added the images generated by the unsupervised DCGAN model to the training set, and the results of the test were statistically analyzed. As can be seen from Table 3, the CM feature has a very good effect on Dog1, and its accuracy rate is 94.99%, and the effect on Dog2, Dog3, and Dog5 is also good, and the accuracy rate of more than 90% can be achieved. The methods proposed in this paper obtained the highest accuracy rates on Dog2, Dog3 and Dog5, which were 98.15%, 95.31% and 97.76%, respectively, and similar to the best method on Dog1, both reaching 94.5%.

Table 3. Comparison results

Method	Dog1	Dog2	Dog3	Dog4	Dog5
DCGAN+ELM	0.7465	0.6855	0.7065	**0.971**	0.75
CNN	0.75	0.7405	0.642	0.753	0.6205
ELM	0.4854	0.5925	0.5033	0.6218	0.5428
CM	**0.9499**	0.9174	0.9511	0.8879	0.9302
SE	0.9475	0.9173	0.9494	0.8915	0.9364
SRP	0.9477	0.9129	0.9519	0.8915	0.9335
Splice	0.9441	0.9112	0.9477	0.8901	0.9362
SS-DCGAN	0.9460	**0.9815**	**0.9531**	0.9190	**0.9776**

5. Conclusion

This paper proposes a semi-supervised deep convolutional generative adversarial nets model to learn and classify epileptic EEG data. The main advantages are:

1) Introducing the generative adversarial nets model into EEG signal recognition model. In the model, we learn deep feature and generate pseudo data using the generator.

2) Sending the unlabeled data and the labeled data are to the discriminator for learning and classification in a semi-supervised way, and reducing the model's dependence on labeled data.

3) optimization between the generator and the discriminator in adversarial way, and adjusting the discriminator loss function, adding an optimization constraint direction to achieve accurate classification effect.

4) The experimental results of this method on each dataset show that the model has a high classification accuracy.

This paper proposes an epilepsy prediction framework based on a semi-supervised deep convolutional generated adversarial network (SS-DCGAN). In the framework, we use the semi-supervised deep convolution generator to generate data that matches the correct data distribution, which makes the deep learning method can achieve better experimental results. In general, there is missing of generation data distribution diversity, which makes the loss function of DCGAN model lack of constraints for the distribution of generated data. In order to enhance the model's constraints for the process of generating data during the data mapping, and make the mapping distance, among data belong to the same category, more compact , this paper proposed an end-to-end deep learning framework based on semi-supervised learning. In the framework, we put constraints on the processing of data generation, and get better effect.

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